High-Dimensional Feature Selection Context ($D \gg 10^3$, e.g., $D = 12{,}600+$ features)
Current Elite Candidate Subset: $\mathbf{x}_e = [x_{e,1}, x_{e,2}, \ldots, x_{e,D}] \in \{0,1\}^D$ **with optimal fitness** $f(\mathbf{x}_e) = 0.0820$

Feature Index	**Feature f_1** (Core Biomarker)	**Feature f_2** (Pure Noise Gene)	**Feature f_3** (Redundant Gene)	**Feature f_4** (Synergistic Pair)	…	**Feature f_D** (Background)
True Contribution	**Strong Positive** (High Relevance)	**Strong Negative** (Noise / Overfitting)	**Negative** (Redundant of f_1)	**Strong Positive** (Synergy with f_1)	…	**Neutral / Zero** (No Correlation)
Current Subset $\mathbf{x}_e$	**1 (Selected)**	**0 (Excluded)**	**0 (Excluded)**	**1 (Selected)**	…	**0 (Excluded)**
Conventional Mutation (Uniform $p_{\text{mut}} = 1/D$)	**Blind Flip: $1 \to 0$ (Mistakenly Pruned)**	**Blind Flip: $0 \to 1$ (Mistakenly Activated)**	$0 \to 0$ (No flip)	$1 \to 1$ (No flip)	…	$0 \to 0$ (No flip)
Physical Search Lesion (The "Why")	**[Lesion 1] Fatal Loss:** Critical biomarker is discarded; accuracy collapses drastically ($\Delta\text{Acc} = -14.2\%$)	**[Lesion 2] Noise Influx:** Noisy gene activated; bloats feature subset and induces severe overfitting	Redundant gene remains unpruned; wastes model budget	Synergy with f_1 is destroyed due to the loss of f_1	…	Vast background remains unguided by fitness memory
Resulting Mutant $\mathbf{x}'$	$\mathbf{x}' = [0,1,0,1,\ldots,0] \in \{0,1\}^D \implies$ **Degraded Fitness:** $f(\mathbf{x}') = 0.2250 \gg f(\mathbf{x}_e) = 0.0820$ **(Destructive Trial: Massive Evaluation Overhead Incurred with Zero Exploration Progress)**					

Theoretical Bottlenecks of Blind Perturbations in Ultra-High-Dimensional Spaces ($D \gg 10^3$)

Bottleneck 1: Infinitesimal Probability of Beneficial Modification

In spaces with $D \geq 10^4$ dimensions, the combinatorial neighborhood is 2^D. An undirected random bit flip has an infinitesimal hit probability $\mathcal{O}(1/D) \approx 10^{-4}$ of modifying the few decisive biomarkers, while having an overwhelming probability $(1 - d^*/D) \approx 99.8\%$ of perturbing irrelevant noise.

Bottleneck 2: Memoryless Search Amnesia

Standard meta-heuristics immediately discard the fitness evaluation feedback of trial solutions after particle updates. The optimizer fails to accumulate historical knowledge regarding which features systematically improved classification accuracy versus which features induced error spikes.

The KDS-EL Theoretical Closed-Loop Solution (Section 3.3)

KDS-EL resolves these dilemmas through a closed loop: cumulative trial feedback is preserved in the continuous importance vector $\mathbf{W} \in \mathbb{R}^D$ with exponential decay $\gamma = 0.99$. The KER operator computes directional scores $\sigma_j = (1 - 2x_{e,j}) \cdot W_j$ to deterministically prune weak features ($1 \to 0$) and add strong features ($0 \to 1$) under bounded computational cost ($k_e \ll N$).